Exno: 12
10/10/26

End-End Communication at Transport layer

Aim: a) Implement echo client server using TCP/UDP sockets.

b) Implement a chat program using socket programming.

Code:

TCP

```
import socket
import threading

# -- server --
def start_server():
    server = socket.socket(socket.AF_INET, socket.
                           SOCK_STREAM)
    server.bind(('127.0.0.1', 5000))
    server.listen(1)
    print("[SERVER] Listening on 127.0.0.1:5000")

    conn, addr = server.accept()
    print(f"[SERVER] Connected by {addr}")

    while True:
        data = conn.recv(1024).decode()
        if not data or data.lower() == 'quit':
            print("[SERVER] Connection closed by client.")
            break
        print(f"[SERVER] Received: {data}")
        conn.send(f"Server received: {data}".encode())

    conn.close()
    server.close()
```

```
# -- CLIENT --
def start_client():
    client = socket.socket(socket.AF_INET, socket.
                           SOCK_STREAM)

    client.connect(('127.0.0.1', 5000))
    print("[CLIENT] Connected to server")

    while True:
        msg = input("Enter message (or 'quit' to exit): ")
        client.send(msg.encode())
        if msg.lower() == 'quit':
            break
        response = client.recv(1024).decode()
        print(f"[SERVER RESPONSE] {response}")
    client.close()

# -- Main --
if __name__ == "__main__":
    # Run server in a background thread
    server_thread = threading.Thread(target=start_
                    server, daemon=True)
    server_thread.start()

    # Run client in main thread
    start_client()
```

Output:

~~=== RESTART: C:/~~

[SERVER] Listening on 127.0.0.1:5000 [CLIENT] Connected to server

[SERVER] Connected by ('127.0.0.1', 53458) Enter message (or 'quit' to exit):

hello

[SERVER] Received: hello

[SERVER RESPONSE] Server received : hello
Enter message (or 'quit' to exit):

ii)

[SERVER] listening on 127.0.0.1: 5000 [CLIENT] Connected to server

[SERVER] Connected by ('127.0.0.1', 60328) Enter message (or 'quit' to exit):

Hi
[SERVER] Received : HI
[SERVER RESPONSE] Server received : HI
Enter message (or 'quit' to exit): quit
[SERVER] Connection closed by client.

UDP

Code:

```
import socket
import threading

# -- SERVER --
def udp_server():
    s = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
    s.bind(('127.0.0.1', 12000))
    print("[SERVER] Running on 127.0.0.1:12000")
    while True:
        data, addr = s.recvfrom(1024)
        msg = data.decode()
        if msg.lower() == 'quit':
            print(f"[SERVER] Client {addr} ended the connection.")
            break
```

```
        print(f"[SERVER] Received from {addr}: {msg}")
        s.sendto(data, addr)  # Echo back
    s.close()
    print("[SERVER] Stopped.")

# -- Client --
def udp_client():
    c = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
    server = ('127.0.0.1', 12000)
    while True:
        msg = input("Enter message (or 'quit' to stop): ")
        c.sendto(msg.encode(), server)
        if msg.lower() == 'quit':
            print("[CLIENT] Closing connection...")
            break
        data, _ = c.recvfrom(1024)
        print("Echo from server:", data.decode())
    c.close()
    print("[CLIENT] Stopped.")

# -- Main --
if __name__ == "__main__":
    threading.Thread(target=udp_server, daemon=True).start()
    udp_client()
```

Output:

```
Enter message (or 'quit' to stop): [SERVER] Running on 127.0.0.1:12000
hi
[SERVER] Received from ('127.0.0.1', 49870): hi
Echo from server: hi
```

Enter message (or 'quit' to stop): hello
[SERVER] Received from ('127.0.0.1', 49670): hello
Echo from server : hello
Enter message (or 'quit' to stop): how are you
[SERVER] Received from ('127.0.0.1', 49670): how are you
Echo from server : how are you
Enter message (or 'quit' to stop): quit
[CLIENT] Closing connection... [SERVER] Client ('127.0.0.1', 49670) ended the connection.

[CLIENT] Stopped. [SERVER] Stopped.